

Third Normal Form (3NF) in DBMS

Transitive Dependency, Formal Rules, GATE Practice & Decision Shortcuts

Third Normal Form (3NF) is one of the most essential normalization concepts in database management systems. It is widely tested in GATE, UGC NET, semester exams, and technical interviews.

1. What is Third Normal Form?

A relation R is in Third Normal Form (3NF) if:

Rule 1: The relation must already be in **Second Normal Form (2NF)**.

Rule 2: There should be **NO Transitive Dependency** of a non-prime attribute on any candidate key.

Formal Exam Definition ★: A relation is in 3NF if for every non-trivial functional dependency $X \rightarrow A$, either X is a Superkey OR A is a Prime Attribute.

2. What is Transitive Dependency?

If $A \rightarrow B$ and $B \rightarrow C$, then by transitivity, $A \rightarrow C$.

Suppose A is a Candidate Key, while B and C are Non-Prime attributes.

Transitive Dependency Pattern ★★ ★:

Candidate Key (A) $\rightarrow$ Non-Prime (B) $\rightarrow$ Non-Prime (C)

Here, C indirectly depends on the candidate key A through another non-prime attribute B . This pattern violates 3NF.

3. Student Example: State & City

Consider table Student(Roll_No, State, City):

Roll_No	State	City
1	Punjab	Mohali
2	Karnataka	Bangalore
3	Punjab	Mohali

Functional Dependencies:

- $Roll_No \rightarrow State$
- $State \rightarrow City$

- **Candidate Key:** *Roll_No*
- **Prime Attribute:** *Roll_No*
- **Non-Prime Attributes:** {*State*, *City*}

Look at *State* → *City*: Both *State* and *City* are non-prime attributes (*ext{Non-Prime}* → *ext{Non-Prime}*). This introduces a transitive dependency (*Roll_No* → *State* → *City*). Hence, the table is **NOT in 3NF**.

4. Formal 3NF Condition ★★☆☆

For every functional dependency $X \rightarrow A$, at least ONE of these conditions must hold true:

$X \text{ ext\{ is a Superkey\}}$ **OR** $A \text{ ext\{ is a Prime Attribute\}}$

- If **either** condition is satisfied, the FD complies with 3NF.
- If **both** conditions fail (X is not a superkey AND A is non-prime), the relation **FAILS 3NF**.

5. The "Prime RHS" Exception Explained

A common trap in exams is assuming that "*any non-prime attribute determining anything violates 3NF*."

If $X \rightarrow A$ exists where X is not a superkey, but A is a **Prime Attribute** (part of some candidate key), this FD is **ALLOWED in 3NF**!

6. 2NF vs. 3NF Comparison

Feature	2NF (Second Normal Form)	3NF (Third Normal Form)
Prerequisite	Must be in 1NF	Must be in 2NF
Target Problem	Partial Dependency	Transitive Dependency
Violation Condition	$LHS \not\subset \text{ext\{Candidate Key\}}$ AND $RHS = \text{ext\{Non-Prime\}}$	$LHS \neq \text{ext\{Superkey\}}$ AND $RHS = \text{ext\{Non-Prime\}}$
Logic Operator	AND (Both true = Violation)	OR (Either true = Satisfied)

7. GATE-Style Practice Problem

Given relation $R(A, B, C, D)$ with FDs:

$AB \rightarrow CD, D \rightarrow A$

Step 1: Find ALL Candidate Keys

- $(AB)^+ = \{A, B, C, D\} \Rightarrow AB$ is a Candidate Key.
- Since $D \rightarrow A$, substitute A with D : $(DB)^+ = \{D, B, A, C\} \Rightarrow DB$ is also a Candidate Key.

Candidate Keys: {*AB*, *DB*}

Step 2: Classify Attributes

- **Prime Attributes:** $\{A, B, D\}$ (participate in at least one candidate key)
- **Non-Prime Attributes:** $\{C\}$

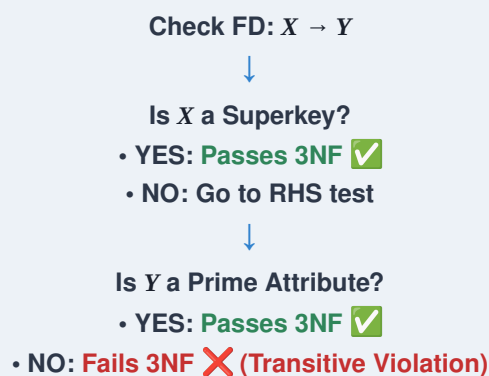
Step 3: Test FDs against 3NF Condition

FD	Condition 1: Is LHS Superkey?	Condition 2: Is RHS Prime?	3NF Status
$AB \rightarrow CD$	YES (AB is CK)	—	SATISFIED ✓
$D \rightarrow A$	NO (D is not superkey)	YES (A is prime)	SATISFIED ✓

Final Answer: The relation R is **IN 3NF**. Notice that $D \rightarrow A$ is allowed because its RHS (A) is a prime attribute!

8. Quick Decision Flowchart for 3NF

For every functional dependency $X \rightarrow Y$ in relation R :



9. Master Summary Cheat Sheet

Normal Form	Core Requirement / Condition
1NF	Atomic values, no repeating groups.
2NF	1NF + No partial dependency ($\text{ext}\{\text{Proper Subset of CK}\} \rightarrow \text{ext}\{\text{Non-Prime}\}$).
3NF	2NF + No transitive dependency (For $X \rightarrow A$, X is Superkey OR A is Prime).
BCNF	For every non-trivial FD $X \rightarrow A$, X MUST be a Superkey (stricter than 3NF).

Golden Exam Tip: Always find *ALL* candidate keys first before checking 2NF or 3NF. Missing even one candidate key can cause you to misclassify prime attributes and mark the wrong answer!